

Supplementary material: Target-conditional information horizons in wind-power event forecasting

This supplement belongs to the Renewable Energy manuscript *Different information horizons shape wind-power event predictability*. It replaces the legacy event-representation supplement and documents the current NP005–NP037 evidence package.

S1. Study object and information boundary

The target begins at T and contains 17 native 15-min power nodes through $T + 4$ h. Pizhou power is divided by the provider-supplied 87.45-MW farm capacity; Suining and the external process sites use their site-specific training-period q99.5 scales. All sites use a 0.2 normalized excursion rule, with the physical scale declared separately. The four state classes are neither, upward, downward and both. The downward first-passage label has 16 crossing bins (15–240 min) and a 17th no-crossing bin. Power history ends before the issue time. The strict-clock weather package uses a value fixed 24 h before the valid target time. Future onsite wind and ERA5 are retrospective diagnostics or matured labels and do not enter an operating issue-time row.

The administrative reference for Pizhou is 34.3403°N, 118.0068°E and the supplied onsite wind reference height is 140 m. The Suining aggregate contains 14 turbines and retains a row when at least 80% of turbines are valid. Each turbine is scaled by its first chronological training-period q99.5 value. No arbitrary height conversion is applied; the weather products retain their source heights and are interpreted as issued product information.

S2. Issue-time contracts and chronological support

S2.1 Pizhou core process-state protocol (NP005/NP008)

- NP005 question: How do fixed-target event risks and matched information gains change as observations become older?
- Pizhou target: 17 native quarter-hour power nodes from T to $T+4$ h; 0.2 of the provider-supplied 87.45-MW farm capacity; power divided by 87.45 MW; no clipping of recorded support values.
- Leads: 15, 60, 120, 240, 480, 720 min; test cohort: 6,496 windows; arms: power, power_weather, power_weather_wind; model: ExtraTrees 150 trees, depth 16, leaf 32; no test selection.
- Clock/split: Each issue is $T-L$; onsite measurements end no later than issue minus 1min; UTC boundaries 2024-07-08 16:00:00+00:00, 2024-10-20 16:00:00+00:00; target support Identical target windows across all leads/arms; earliest issue has 24h split purge, target end inside split.

- Maturity and update: $T+4h+1min \leq \text{UTC midnight update} \leq \text{issue}$; preceding 56day target starts; NP008 reuses frozen NP007 rolling predictions and fixes lambda 0.001 before test scoring.
- NP008 features: `issue_power`, `issue_power_squared`, `GFS_delta/5`, `JMA_delta/5`, `GFS_mean/20`, `GFS_range/10`, `issue_power*GFS_delta/5`, `abs_GFS_delta/5`, `abs_JMA_delta/5`, `GFS_coarse_excess_TV/10`; state classes: `Unchanged17node` future4h path and four event classes; paired 6496 test windows; source hashes and freeze receipt are stored in `outputs/next_paper/np008/protocol.json` and `freeze.json`.

S2.2 Pizhou timing protocol (NP009)

- Question: Does forecast temporal ordering improve arrival-time allocation beyond occurrence risk and background?
- Timing bins: First ordered `drawup/down >= 0.2capacity` within `unchanged17-point[T, T+4h]` target; bins 15..240min and none; occurrence: Frozen NP008 full rolling four-class marginals; same p for every timing method; arms: frequency, context, ordered, sorted; feature counts: `{"context": 14, "ordered": 44, "sorted": 44}`; lambda selection: One lambda minimizes mean positive-event conditional RPS across 2 directions 6 leads 3 covariate arms on validation.
- Arm construction: context uses endpoint/context covariates; ordered uses the ordered NWP interior trajectory; sorted independently sorts the same interior values and preserves endpoint/context features. Fitting: Only matured positive windows; 16-time multinomial with mean-loss $L2, C=1/(n*\lambda)$; same causal 56day rule; smoothing: Predictive mixture with uniform mass $16/(n+16)$; empirical frequency adds 1 to each bin; <100 positive or <3 classes fallback.
- The validation-frozen timing policy selects $\lambda=0.001$ before test scoring; `selection.json` records the frozen policy and validation hash. NP009 is a 16-time multinomial timing model, not the ExtraTrees timing model used by NP025/NP028.
- NP009 uses the frozen NP008 full rolling occurrence marginals. At 15 min, the downward timing denominator is 2,339 windows; frequency conditional RPS/median-MAE are 0.161377/54.215 min and context values are 0.148245/49.348 min.

S2.3 Pizhou headline absolute scores

package	lead_min	arm	metric	score	n
NP008	720	frequency	return	0.135149	6496
		rolling	Brier		
NP008	720	full rolling	return	0.117423	6496
			Brier		

package	lead_min	arm	metric	score	n
NP008	720	frequency	multiclass	0.706319	6496
NP008	720	rolling	Brier		
NP008	720	full rolling	multiclass	0.572285	6496
NP009	15	frequency	Brier		
NP009	15	frequency	conditional	0.161377	2339
NP009	15	frequency	timing		
NP009	15	frequency	RPS		
NP009	15	context	conditional	0.148245	2339
NP009	15	context	timing		
NP009	15	context	RPS		
NP009	15	frequency	conditional	54.215477	2339
NP009	15	frequency	median		
NP009	15	frequency	MAE		
NP009	15	frequency	(min)		
NP009	15	context	conditional	49.348012	2339
NP009	15	context	median		
NP009	15	context	MAE		
NP009	15	context	(min)		

NP025: THREE_ARM_INFORMATION_DECOMPOSITION - Question: Which process objects receive information from issue-time power history and external historical-forecast weather when each source is evaluated alone and together? - Site and period: suining; 2023-08-31 16:00:00+00:00 to 2024-08-31 15:45:00+00:00. - Leads: 15, 60, 120, 240, 480, 720 min; target: 17 native 15-min nodes; 0.2 excursion; four process classes and downward first passage. - Arms: {"power": "16 issue-time power lags plus known target calendar", "weather": "known target calendar plus four weather variables interpolated to 17 target nodes", "joint": "power arm plus weather arm"}. - Split: Chronological 60/20/20 blocks with four-hour purge; fixed model budgets; no test selection; event model: ExtraTrees 150 trees, depth16, leaf32, max_features=1.0, seed41; timing model: ExtraTrees 100 trees, depth16, leaf16, max_features=1.0, seed43; occurrence held at training frequency. - Weather source: Open-Meteo Historical Forecast API; GFS global; manifest and monthly response hashes retained; variables: u100_ms, v100_ms, temperature_1000hPa, surface_pressure; complete support: all requested rows in the archived combined file.

NP028: STRICT_ISSUE_TIME_PREVIOUS_DAY1_WEATHER - Question: Does a fixed previous-day1 external GFS package provide process-state information under a strict issue-time contract? - Site and period: suining; 2023-08-31 16:00:00+00:00 to 2024-08-31 15:45:00+00:00. - Leads: 15, 60, 120, 240, 480, 720 min; target: 17 native 15-min nodes; 0.2 excursion; four classes and downward first passage. - Arms: {"power": "16 issue-time power lags plus target calendar", "weather": "target calendar plus previous_day1 u100/v100, surface pressure, and 2m temperature", "joint": "power arm plus previous_day1

weather arm”}. - Split: Chronological 60/20/20 support-period blocks with four-hour purge; fixed budgets; no test selection; event model: ExtraTrees 150 trees, depth16, leaf32, max_features=1.0, seed41; timing model: ExtraTrees 100 trees, depth16, leaf16, max_features=1.0, seed43; occurrence held at training frequency. - Weather source: Open-Meteo Previous Runs API; GFS global; all weather values are fixed 24 h before valid time; variables: u100_ms, v100_ms, surface_pressure_previous_day1, temperature_2m_previous_day1; complete support: 2024-02-16 06:00:00+00:00 to 2024-08-31 15:45:00+00:00.

For NP025, the historical-forecast trajectory is the Open-Meteo GFS global historical-forecast package. Each row is selected by the target-valid trajectory and its archived product timestamp; the row-level rule is that all requested fields must be present in the historical package before the forecast row is accepted. The archive does not expose a model release timestamp, so NP025 is reported as a retrospective information decomposition and is not used as the strict real-time release test. NP028 uses the Open-Meteo Previous Runs GFS global package and fixes every weather value 24 h before target valid time. Complete NP028 weather support begins on 2024-02-16. The NP025 and NP028 weather manifests retain monthly query URLs, response hashes, coordinates, units and returned timestamps.

The chronological split is 60/20/20 with a four-hour purge at each boundary. The event model is ExtraTrees with 150 trees, maximum depth 16, minimum leaf size 32, all available features and seed 41. The conditional timing model uses ExtraTrees with 100 trees, depth 16, leaf size 16 and seed 43 on matured downward-event labels; the occurrence probability is held at the training frequency for the conditional timing product. The matched arms use identical windows and budgets within each package.

S3. Label and score definitions

For a target path Y_T , the upward and downward excursions are running-extrema functionals. A downward crossing label $\tau_{T,\downarrow}$ is the first 15-min node at which the downward excursion reaches 0.2 on the declared site-specific normalized scale; the no-crossing label is bin 16. Conditional timing RPS and median arrival-time error are evaluated on the crossing subset, whose denominator is reported in every table. Event-state Brier scores use all test windows.

The unconditional joint first-passage distribution used as a diagnostic is formed from the event probability $p_{\downarrow} = p_{\text{downward}} + p_{\text{both}}$ and the conditional timing distribution q :

$$\tilde{q}_{0:15} = p_{\downarrow} q_{0:15}, \quad \tilde{q}_{16} = 1 - p_{\downarrow}.$$

The joint first-passage RPS is reported as an accounting score that retains the occurrence and timing components in one no-crossing-inclusive distribution. The

main paper keeps conditional RPS as the primary timing object because the route is designed to improve the timing distribution given a process occurrence.

Relative loss reduction is $100(1 - L_{\text{candidate}}/L_{\text{reference}})$. Intervals are paired seven-day calendar-block bootstrap percentile intervals with 2,000 draws and seed 41; the same target blocks are used for the compared arms. The fitted models are held fixed in these intervals.

S4. External power-history process-object validation (NP022)

NP022 uses 15-min SCADA from La Haute Borne (4 turbines, 13,912 test windows and 3,966 downward events at 15 min; 2014–2015) and Suining (14 turbines, 6,996 test windows and 2,653 downward events at 15 min; 2023–2024), with a 17-node, four-hour target and the same normalized excursion rule on each site’s training-period q99.5 scale. Each site uses 16 power-history lags ending at issue time, known calendar terms, chronological 60/20/20 splits and a four-hour purge. The result validates the occurrence-versus-first-passage distinction for a power-history forecast object; it does not validate the weather-source increment. Suining is reused as the site for NP024–NP030 information-arm experiments, while La Haute Borne supplies the independent external farm check. The exact 15-min timing results are shown below; intervals are paired seven-day calendar-block bootstrap intervals.

site	lead_min	metric	relative_loss	reduction	high_pct	windows	valid_windows
lahaute	15	conditional	6.914	3.694	10.149	13912	3966
lahaute	15	conditional	7.655	4.099	11.257	13912	3966
suining	15	conditional	6.625	3.476	9.058	6996	2653
suining	15	conditional	7.644	4.506	10.087	6996	2653

S5. Three-arm information decomposition (NP025)

The three matched arms are power history plus calendar, weather trajectory plus calendar, and their concatenation. The following tables give absolute scores, denominators and the complete six-lead matrix.

S5.1 Absolute scores

package	lead_min	arm	metric	score	n
np025	15	power	multiclass Brier	0.575612	6996
np025	15	power	return Brier	0.160101	6996
np025	15	power	conditional timing RPS	0.154844	2653
np025	15	power	joint first-passage RPS	1.703315	6996
np025	15	weather	multiclass Brier	0.623027	6996

package	lead_min	arm	metric	score	n
np025	15	weather	return Brier	0.174254	6996
np025	15	weather	conditional timing RPS	0.170936	2653
np025	15	weather	joint first-passage RPS	2.252539	6996
np025	15	joint	multiclass Brier	0.561538	6996
np025	15	joint	return Brier	0.164498	6996
np025	15	joint	conditional timing RPS	0.153557	2653
np025	15	joint	joint first-passage RPS	1.669583	6996
np025	60	power	multiclass Brier	0.600090	6996
np025	60	power	return Brier	0.164154	6996
np025	60	power	conditional timing RPS	0.161427	2653
np025	60	power	joint first-passage RPS	1.887056	6996
np025	60	weather	multiclass Brier	0.623339	6996
np025	60	weather	return Brier	0.174356	6996
np025	60	weather	conditional timing RPS	0.170936	2653
np025	60	weather	joint first-passage RPS	2.252990	6996
np025	60	joint	multiclass Brier	0.583377	6996
np025	60	joint	return Brier	0.166401	6996
np025	60	joint	conditional timing RPS	0.160125	2653
np025	60	joint	joint first-passage RPS	1.881277	6996
np025	120	power	multiclass Brier	0.615617	6996
np025	120	power	return Brier	0.166580	6996
np025	120	power	conditional timing RPS	0.164451	2653
np025	120	power	joint first-passage RPS	2.004923	6996
np025	120	weather	multiclass Brier	0.623467	6996
np025	120	weather	return Brier	0.174329	6996
np025	120	weather	conditional timing RPS	0.170936	2653
np025	120	weather	joint first-passage RPS	2.253479	6996
np025	120	joint	multiclass Brier	0.592563	6996
np025	120	joint	return Brier	0.165971	6996
np025	120	joint	conditional timing RPS	0.167480	2653
np025	120	joint	joint first-passage RPS	2.013336	6996
np025	240	power	multiclass Brier	0.644034	6996
np025	240	power	return Brier	0.173388	6996
np025	240	power	conditional timing RPS	0.166670	2653
np025	240	power	joint first-passage RPS	2.145147	6996
np025	240	weather	multiclass Brier	0.623419	6996
np025	240	weather	return Brier	0.174217	6996
np025	240	weather	conditional timing RPS	0.170936	2653
np025	240	weather	joint first-passage RPS	2.251818	6996
np025	240	joint	multiclass Brier	0.610165	6996
np025	240	joint	return Brier	0.172187	6996
np025	240	joint	conditional timing RPS	0.182556	2653
np025	240	joint	joint first-passage RPS	2.166719	6996
np025	480	power	multiclass Brier	0.670330	6996

package	lead_min	arm	metric	score	n
np025	480	power	return Brier	0.179329	6996
np025	480	power	conditional timing RPS	0.166551	2653
np025	480	power	joint first-passage RPS	2.308316	6996
np025	480	weather	multiclass Brier	0.622916	6996
np025	480	weather	return Brier	0.173840	6996
np025	480	weather	conditional timing RPS	0.170936	2653
np025	480	weather	joint first-passage RPS	2.252441	6996
np025	480	joint	multiclass Brier	0.624551	6996
np025	480	joint	return Brier	0.174619	6996
np025	480	joint	conditional timing RPS	0.182326	2653
np025	480	joint	joint first-passage RPS	2.241706	6996
np025	720	power	multiclass Brier	0.688773	6996
np025	720	power	return Brier	0.181834	6996
np025	720	power	conditional timing RPS	0.169396	2653
np025	720	power	joint first-passage RPS	2.388272	6996
np025	720	weather	multiclass Brier	0.622417	6996
np025	720	weather	return Brier	0.173585	6996
np025	720	weather	conditional timing RPS	0.170936	2653
np025	720	weather	joint first-passage RPS	2.247789	6996
np025	720	joint	multiclass Brier	0.621429	6996
np025	720	joint	return Brier	0.174678	6996
np025	720	joint	conditional timing RPS	0.168204	2653
np025	720	joint	joint first-passage RPS	2.230083	6996
np028	15	power	multiclass Brier	0.619739	3772
np028	15	power	return Brier	0.161253	3772
np028	15	power	conditional timing RPS	0.153839	1328
np028	15	power	joint first-passage RPS	1.716899	3772
np028	15	weather	multiclass Brier	0.677309	3772
np028	15	weather	return Brier	0.159760	3772
np028	15	weather	conditional timing RPS	0.168910	1328
np028	15	weather	joint first-passage RPS	2.327551	3772
np028	15	joint	multiclass Brier	0.583617	3772
np028	15	joint	return Brier	0.154105	3772
np028	15	joint	conditional timing RPS	0.155691	1328
np028	15	joint	joint first-passage RPS	1.714607	3772
np028	60	power	multiclass Brier	0.652818	3772
np028	60	power	return Brier	0.163466	3772
np028	60	power	conditional timing RPS	0.159858	1328
np028	60	power	joint first-passage RPS	1.894041	3772
np028	60	weather	multiclass Brier	0.676935	3772
np028	60	weather	return Brier	0.159343	3772
np028	60	weather	conditional timing RPS	0.168910	1328
np028	60	weather	joint first-passage RPS	2.321699	3772
np028	60	joint	multiclass Brier	0.608165	3772

package	lead_min	arm	metric	score	n
np028	60	joint	return Brier	0.156432	3772
np028	60	joint	conditional timing RPS	0.163288	1328
np028	60	joint	joint first-passage RPS	1.887661	3772
np028	120	power	multiclass Brier	0.677774	3772
np028	120	power	return Brier	0.166193	3772
np028	120	power	conditional timing RPS	0.162954	1328
np028	120	power	joint first-passage RPS	2.024545	3772
np028	120	weather	multiclass Brier	0.674381	3772
np028	120	weather	return Brier	0.159339	3772
np028	120	weather	conditional timing RPS	0.168910	1328
np028	120	weather	joint first-passage RPS	2.302440	3772
np028	120	joint	multiclass Brier	0.616242	3772
np028	120	joint	return Brier	0.154848	3772
np028	120	joint	conditional timing RPS	0.165522	1328
np028	120	joint	joint first-passage RPS	1.971873	3772
np028	240	power	multiclass Brier	0.700688	3772
np028	240	power	return Brier	0.172616	3772
np028	240	power	conditional timing RPS	0.167666	1328
np028	240	power	joint first-passage RPS	2.184391	3772
np028	240	weather	multiclass Brier	0.670590	3772
np028	240	weather	return Brier	0.159633	3772
np028	240	weather	conditional timing RPS	0.169272	1328
np028	240	weather	joint first-passage RPS	2.287425	3772
np028	240	joint	multiclass Brier	0.632775	3772
np028	240	joint	return Brier	0.156651	3772
np028	240	joint	conditional timing RPS	0.168180	1328
np028	240	joint	joint first-passage RPS	2.053378	3772
np028	480	power	multiclass Brier	0.749283	3772
np028	480	power	return Brier	0.175866	3772
np028	480	power	conditional timing RPS	0.167251	1328
np028	480	power	joint first-passage RPS	2.539761	3772
np028	480	weather	multiclass Brier	0.675134	3772
np028	480	weather	return Brier	0.159812	3772
np028	480	weather	conditional timing RPS	0.168978	1328
np028	480	weather	joint first-passage RPS	2.293023	3772
np028	480	joint	multiclass Brier	0.656977	3772
np028	480	joint	return Brier	0.159748	3772
np028	480	joint	conditional timing RPS	0.168577	1328
np028	480	joint	joint first-passage RPS	2.206822	3772
np028	720	power	multiclass Brier	0.768864	3772
np028	720	power	return Brier	0.177386	3772
np028	720	power	conditional timing RPS	0.161916	1328
np028	720	power	joint first-passage RPS	2.639331	3772
np028	720	weather	multiclass Brier	0.675779	3772

package	lead_min	arm	metric	score	n
np028	720	weather	return Brier	0.160541	3772
np028	720	weather	conditional timing RPS	0.169256	1328
np028	720	weather	joint first-passage RPS	2.313774	3772
np028	720	joint	multiclass Brier	0.672560	3772
np028	720	joint	return Brier	0.161167	3772
np028	720	joint	conditional timing RPS	0.166304	1328
np028	720	joint	joint first-passage RPS	2.299966	3772

S5.2 No-crossing-inclusive joint accounting

`joint_RPS` is the 17-bin score on all windows. `conditional_RPS_positive` is the original positive-event conditional score and uses `down_events` as its denominator; it is not an all-window score.

package	lead_min	arm	joint_RPS	event_Brier	conditional_RPS	all_RPS	positive_events
np025	15	power	1.703315	0.575612	0.154844	6996	2653
np025	15	weather	2.252539	0.623027	0.170936	6996	2653
np025	15	joint	1.669583	0.561538	0.153557	6996	2653
np025	60	power	1.887056	0.600090	0.161427	6996	2653
np025	60	weather	2.252990	0.623339	0.170936	6996	2653
np025	60	joint	1.881277	0.583377	0.160125	6996	2653
np025	120	power	2.004923	0.615617	0.164451	6996	2653
np025	120	weather	2.253479	0.623467	0.170936	6996	2653
np025	120	joint	2.013336	0.592563	0.167480	6996	2653
np025	240	power	2.145147	0.644034	0.166670	6996	2653
np025	240	weather	2.251818	0.623419	0.170936	6996	2653
np025	240	joint	2.166719	0.610165	0.182556	6996	2653
np025	480	power	2.308316	0.670330	0.166551	6996	2653
np025	480	weather	2.252441	0.622916	0.170936	6996	2653
np025	480	joint	2.241706	0.624551	0.182326	6996	2653
np025	720	power	2.388272	0.688773	0.169396	6996	2653
np025	720	weather	2.247789	0.622417	0.170936	6996	2653
np025	720	joint	2.230083	0.621429	0.168204	6996	2653
np028	15	power	1.716899	0.619739	0.153839	3772	1328
np028	15	weather	2.327551	0.677309	0.168910	3772	1328
np028	15	joint	1.714607	0.583617	0.155691	3772	1328
np028	60	power	1.894041	0.652818	0.159858	3772	1328
np028	60	weather	2.321699	0.676935	0.168910	3772	1328
np028	60	joint	1.887661	0.608165	0.163288	3772	1328
np028	120	power	2.024545	0.677774	0.162954	3772	1328
np028	120	weather	2.302440	0.674381	0.168910	3772	1328
np028	120	joint	1.971873	0.616242	0.165522	3772	1328
np028	240	power	2.184391	0.700688	0.167666	3772	1328

package	lead_min	arm	joint_RPS	event_Brier	conditional_RPS	all_RPS	positive events
np028	240	weather	2.287425	0.670590	0.169272	3772	1328
np028	240	joint	2.053378	0.632775	0.168180	3772	1328
np028	480	power	2.539761	0.749283	0.167251	3772	1328
np028	480	weather	2.293023	0.675134	0.168978	3772	1328
np028	480	joint	2.206822	0.656977	0.168577	3772	1328
np028	720	power	2.639331	0.768864	0.161916	3772	1328
np028	720	weather	2.313774	0.675779	0.169256	3772	1328
np028	720	joint	2.299966	0.672560	0.166304	3772	1328

S6. Calendar-only information control (NP031)

NP031 fits the same ExtraTrees event and timing budgets on only the four known target-calendar variables extracted from the frozen NP025 and NP028 datasets. It uses the same train/test masks, target rows and seven-day paired bootstrap. The control is deliberately reported against the frequency reference: it shows whether the headline increments require power or weather information beyond calendar seasonality.

package	lead_min	task	metric	relative_low	in_high	calendar	frequency	reference_score
np025	15	event	return_brier	-	-	0.2071	0.1814	6996
				14.1702	17.9079	8.2785		
np025	15	event	multiclass_brier	-	-	0.8421	0.6922	6996
				21.6583	30.7799	9.0857		
np025	15	downward	conditional_rps	-	-	0.1708	0.1658	6996
				2.9968	4.8543	1.2232		
np025	15	downward	conditional_median_mae_minutes	-	-	55.5971	55.1772	6996
				4.3857	6.8077	1.7052		
np025	60	event	return_brier	-	-	0.2070	0.1814	6996
				14.0702	17.7205	8.2810		
np025	60	event	multiclass_brier	-	-	0.8378	0.6922	6996
				21.0482	30.0026	8.6542		
np025	60	downward	conditional_rps	-	-	0.1708	0.1658	6996
				2.9968	4.8543	1.2232		
np025	60	downward	conditional_median_mae_minutes	-	-	55.5971	55.1772	6996
				4.3857	6.8077	1.7052		
np025	120	event	return_brier	-	-	0.2075	0.1814	6996
				14.3417	18.0079	8.5330		
np025	120	event	multiclass_brier	-	-	0.8386	0.6921	6996
				21.1662	30.1581	8.6800		
np025	120	downward	conditional_rps	-	-	0.1708	0.1658	6996
				2.9968	4.8543	1.2232		
np025	120	downward	conditional_median_mae_minutes	-	-	55.5971	55.1772	6996
				4.3857	6.8077	1.7052		

package	lead	task	metric	relative_low	in_high	calendar_frequency	reference_score
np025	240	event	return_brier	-	-	0.2076	0.1814 6996
				14.4278	18.1697 8.5476		
np025	240	event	multiclass_brier	-	-	0.8381	0.6921 6996
				21.0943	30.0818 8.6728		
np025	240	downward	conditional_rps	-	-	0.1708	0.1658 6996
				2.9968	4.8543 1.2232		
np025	240	downward	conditional_median_mae_minutes	-	-	55.5971	55.1772 6996
				4.3857	6.8077 1.7052		
np025	480	event	return_brier	-	-	0.2087	0.1814 6996
				15.0350	18.8706 9.0295		
np025	480	event	multiclass_brier	-	-	0.8420	0.6921 6996
				21.6593	30.7796 9.1520		
np025	480	downward	conditional_rps	-	-	0.1708	0.1658 6996
				2.9968	4.8543 1.2232		
np025	480	downward	conditional_median_mae_minutes	-	-	55.5971	55.1772 6996
				4.3857	6.8077 1.7052		
np025	720	event	return_brier	-	-	0.2090	0.1814 6996
				15.2267	19.0535 9.1325		
np025	720	event	multiclass_brier	-	-	0.8408	0.6921 6996
				21.4973	30.6949 8.9407		
np025	720	downward	conditional_rps	-	-	0.1708	0.1658 6996
				2.9968	4.8543 1.2232		
np025	720	downward	conditional_median_mae_minutes	-	-	55.5971	55.1772 6996
				4.3857	6.8077 1.7052		
np028	15	event	return_brier	-	-	0.1799	0.1657 3772
				8.5677	15.5932 1.8888		
np028	15	event	multiclass_brier	-	-	0.8058	0.6787 3772
				18.7265	24.2332 13.2029		
np028	15	downward	conditional_rps	-	-	2.2674	0.1680 0.1658 3772
				1.3328	3.5225		
np028	15	downward	conditional_median_mae_minutes	-	-	56.2613	55.2899 3772
				1.7569	3.9515		
np028	60	event	return_brier	-	-	0.1799	0.1657 3772
				8.5737	15.3728 1.9854		
np028	60	event	multiclass_brier	-	-	0.8118	0.6787 3772
				19.6064	25.4691 14.0056		
np028	60	downward	conditional_rps	-	-	2.2674	0.1680 0.1658 3772
				1.3328	3.5225		
np028	60	downward	conditional_median_mae_minutes	-	-	56.2613	55.2899 3772
				1.7569	3.9515		
np028	120	event	return_brier	-	-	0.1797	0.1657 3772
				8.4323	15.2420 1.7417		
np028	120	event	multiclass_brier	-	-	0.8131	0.6788 3772
				19.7892	25.6869 14.2221		

package	lead	task	metric	relative_low	in_high	calendar	frequency	reference_score
np028	120	downward	conditional_rps -	1.3328	3.5225	2.2674	0.1680	0.1658 3772
np028	120	downward	conditional_median_mae -	1.7569	3.9515	1.9211	0.1680	0.1658 3772
np028	240	event	return_brier -	8.2671	14.9434	1.4807	0.1794	0.1657 3772
np028	240	event	multiclass_brier -	19.5303	25.3454	14.0686	0.8112	0.6787 3772
np028	240	downward	conditional_rps -	1.9251	3.9863	1.5465	0.1690	0.1658 3772
np028	240	downward	conditional_median_mae -	2.2063	4.5195	1.8440	0.1690	0.1658 3772
np028	480	event	return_brier -	8.3645	15.0467	1.7860	0.1796	0.1657 3772
np028	480	event	multiclass_brier -	19.1795	25.0394	13.6208	0.8089	0.6787 3772
np028	480	downward	conditional_rps -	1.6869	3.6548	1.6676	0.1686	0.1658 3772
np028	480	downward	conditional_median_mae -	1.9408	4.1447	1.6884	0.1686	0.1658 3772
np028	720	event	return_brier -	8.5053	14.9592	2.0278	0.1798	0.1657 3772
np028	720	event	multiclass_brier -	20.3707	25.8396	14.4917	0.8167	0.6785 3772
np028	720	downward	conditional_rps -	1.4823	3.4320	1.7714	0.1683	0.1658 3772
np028	720	downward	conditional_median_mae -	1.9408	4.1659	1.6173	0.1683	0.1658 3772

The calendar-only control has negative event-state relative reductions for both packages and negative conditional timing reductions at the primary leads. The information-arm gains therefore include information beyond the shared calendar columns.

S7. Target-aligned routing (NP026 and NP029)

NP026 and NP029 are deterministic transformations of frozen test predictions. The route keeps the joint event-state probabilities and assigns the power-history timing probabilities to downward first passage. No model is refit, no test lead is selected, and no test month is used to define the mapping. State identity is checked by maximum absolute probability difference.

package	lead_minutes	metric	gain (95% CI) %	windows
np026	15	timing RPS	-0.84 (-3.02, 1.19)	6996
np026	15	timing median MAE	-1.11 (-3.18, 0.79)	6996
np026	60	timing RPS	-0.81 (-2.42, 0.75)	6996
np026	60	timing median MAE	-1.07 (-2.69, 0.58)	6996
np026	120	timing RPS	1.81 (-0.18, 3.80)	6996
np026	120	timing median MAE	2.09 (0.10, 4.21)	6996
np026	240	timing RPS	8.70 (4.96, 12.84)	6996
np026	240	timing median MAE	8.01 (5.10, 10.97)	6996
np026	480	timing RPS	8.65 (5.55, 12.20)	6996
np026	480	timing median MAE	8.79 (5.00, 12.94)	6996
np026	720	timing RPS	-0.71 (-2.15, 0.52)	6996
np026	720	timing median MAE	-1.45 (-3.48, 0.66)	6996
np029	15	timing RPS	1.19 (-1.89, 3.60)	3772
np029	15	timing median MAE	2.43 (-1.49, 5.52)	3772
np029	60	timing RPS	2.10 (0.44, 4.15)	3772
np029	60	timing median MAE	2.52 (-0.56, 5.46)	3772
np029	120	timing RPS	1.55 (0.45, 2.23)	3772
np029	120	timing median MAE	1.64 (0.31, 3.14)	3772
np029	240	timing RPS	0.31 (-1.46, 1.61)	3772
np029	240	timing median MAE	-0.22 (-2.04, 0.99)	3772
np029	480	timing RPS	0.79 (-1.39, 4.03)	3772
np029	480	timing median MAE	1.27 (-1.46, 4.96)	3772
np029	720	timing RPS	2.64 (-0.71, 6.42)	3772
np029	720	timing median MAE	2.39 (-1.15, 6.95)	3772

The complete matrices are retained in `outputs/next_paper/np026/routing_summary.csv` and `outputs/next_paper/np029/routing_summary.csv`. The primary route endpoints are 240 and 480 min for the historical package and 60 and 120 min for the strict-clock package; all six leads remain in the supplement.

S8. Monthly stability audits (NP027 and NP030)

NP027 recomputes the frozen NP026 route separately for UTC June, July and August 2024. NP030 recomputes the strict-clock route for July and August 2024. No month, model or lead is selected during the audit. The complete monthly tables, including mixed signs at secondary leads, are retained in the corresponding output directories and figure source manifests. The main text reports the pre-specified supported endpoints; the audit is an auxiliary stability check.

S9. Reproducibility and source traceability

The current evidence package is generated from the following frozen output directories: np005, np007, np008, np009, np013, np014, np016, np017,

np018, np019, np020, np022, np024, np025, np026, np027, np028, np029, np030, np031_calendar_control, np032_np033, np034_weather_ablation, np035_calibration, np036_joint_timing_baseline and np037_threshold_sensitivity.

Each directory contains a protocol or manifest, test predictions or summary, and an independent verification record. NP005/NP007/NP008/NP009 bind the Pizhou cohort, feature definitions, frozen occurrence predictions, timing predictions and validation-frozen lambda. NP013/NP014/NP016–NP020 are retained supporting range/calibration/fusion outputs referenced by the main text and their protocols remain in the corresponding output directories. NP025 and NP028 bind their scripts, SCADA source, weather file and weather manifest by SHA-256. NP026 and NP029 bind their routing scripts to the frozen source prediction manifests. NP031 records the dataset source and the four-column calendar extraction. NP032/NP033 record the seed and calendar robustness extensions; NP034/NP035 record the weather-variable ablation and calibration diagnostics.

NP037 adds the threshold reconstruction check under outputs/next_paper/np037_threshold_sensitivity, and the exact NP022 external timing rows and NP014 range/path rows are reproduced in S4 and S15.

Raw private SCADA is not redistributed. The minimum reproducibility index consists of the protocol JSON, support CSV, prediction summary CSV, verification JSON and source-hash manifest in each listed output directory; these files are sufficient to replay the reported score arithmetic after the private source is replaced by the permitted derived arrays. The released replication package will include public weather files or source URLs, feature/label arrays after privacy review, prediction summaries, source hashes and the exact scripts used to produce the tables. All timestamp handling is UTC.

S10. Secondary matrices and scope

The full six-lead, arm-by-metric relative-loss matrices, all monthly rows, support counts and negative or mixed increments remain part of the machine-readable output directories. The main text uses the pre-specified positive endpoints to explain the information-to-target rule; these secondary matrices provide the complete audit trail for the interpretation.

S11. Seed and month robustness extensions (NP032/NP033)

NP032 refits the NP025 and NP028 event and timing models at seeds 41, 42 and 43 for the 15-, 240- and 720-min endpoints. The central state increments remain stable: at 720 min, joint-versus-power multiclass Brier gains average 9.56% (SD 0.48%) for NP025 and 12.61% (SD 0.57%) for NP028. The full seed matrix is retained below.

index	package	lead_minutes	metric	mean	std	min	max	count
0	np025	15	joint_addition_brier	multiclass_brier 3.356	0.153	2.180	2.445	3
1	np025	15	power_addition_brier	multiclass_brier 8.089	0.417	8.693	9.466	3
2	np025	15	weather_multiclass_brier	8.481	0.227	-	-	3
3	np025	240	joint_addition_brier	multiclass_brier 5.179	0.073	5.116	5.259	3
4	np025	240	power_addition_brier	multiclass_brier 10.038	0.566	1.563	2.665	3
5	np025	240	weather_multiclass_brier	2.971	0.253	2.699	3.201	3
6	np025	720	joint_addition_brier	multiclass_brier 9.555	0.476	9.008	9.880	3
7	np025	720	power_addition_brier	multiclass_brier 10.280	0.879	-	1.239	3
8	np025	720	weather_multiclass_brier	9.368	0.544	0.487	8.742	3
9	np028	15	joint_addition_brier	multiclass_brier 5.738	0.140	5.576	5.828	3
10	np028	15	power_addition_brier	multiclass_brier 10.106	0.356	8.728	9.435	3
11	np028	15	weather_multiclass_brier	9.254	0.930	-	-	3
12	np028	240	joint_addition_brier	multiclass_brier 4.196	0.535	9.692	10.757	3
13	np028	240	power_addition_brier	multiclass_brier 10.039	0.639	0.473	1.667	3
14	np028	240	weather_multiclass_brier	4.241	0.801	3.414	5.014	3
15	np028	720	joint_addition_brier	multiclass_brier 4.612	0.569	12.091	13.219	3
16	np028	720	power_addition_brier	multiclass_brier 10.086	0.122	3.878	4.118	3
17	np028	720	weather_multiclass_brier	4.649	0.528	12.107	13.162	3

NP033 recalculates frozen information increments by calendar month for all six leads. The summary below reports mean, SD, minimum, maximum and count across supported months; the complete row matrix is stored in `outputs/next_paper/np032_np033/monthly_information.csv`.

index	package	arm	task	metric	mean	std	min	max	count
0	np025	joint	event	multiclass_brier	4.345	4.090	7.249	23.241	18
1	np025	power	event	multiclass_brier	8.260	6.511	-	17.568	18
2	np025	power	addition_brier	multiclass_brier 10.767	3.597	3.355	-	10.950	18
3	np025	weather	event	multiclass_brier	4.845	2.657	2.325	7.154	18
4	np028	joint	event	multiclass_brier	6.049	5.177	-	16.904	12
5	np028	power	event	multiclass_brier	7.615	7.615	1.360	10.788	12
6	np028	power	addition_brier	multiclass_brier 10.293	3.692	3.012	-	13.701	12
7	np028	weather	event	multiclass_brier	1.307	0.054	2.884	9.631	12
							-	1.865	12
							1.532		

S12. Calibration diagnostics (NP035)

NP035 bins frozen state probabilities and the probability of a downward arrival within 60 min into ten probability bins. It reports bin count, mean predicted probability, observed frequency and absolute calibration gap for NP025/NP028, all leads and all arms. These diagnostics separate information value from calibration: a lower forecast loss can coexist with a calibration gap, and the main paper therefore uses proper losses as its primary comparison while retaining reliability rows here.

index	package	target	mean	max	count
0	np025	downward	0.0892	0.9058	171
1	np025	downward_arrival_le_60min	0.1891	0.6274	121
2	np025	return	0.1060	0.4032	109
3	np028	downward	0.1124	0.4749	169
4	np028	downward_arrival_le_60min	0.1850	0.6097	112
5	np028	return	0.0821	0.4398	121

The representative bin-level rows below expose the calibration calculation for the 15-min and 12-h joint arms; the complete reliability table is stored in `outputs/next_paper/np035_calibration/reliability.csv`.

package	lead_min	target	bin_low	bin_high	count	mean_preds	obs_rate	abs_gap
np025	15	return	0.0000	0.1000	3069	0.0392	0.0997	0.0605
np025	15	return	0.1000	0.2000	1387	0.1501	0.2408	0.0907
np025	15	return	0.2000	0.3000	1178	0.2488	0.3438	0.0950
np025	15	return	0.3000	0.4000	872	0.3464	0.3945	0.0481
np025	15	return	0.4000	0.5000	425	0.4347	0.4188	0.0159
np025	15	return	0.5000	0.6000	45	0.5361	0.8889	0.3528
np025	15	return	0.6000	0.7000	20	0.6328	0.7500	0.1172
np025	15	downward	0.0000	0.1000	2364	0.0402	0.0943	0.0541
np025	15	downward	0.1000	0.2000	850	0.1480	0.2282	0.0803
np025	15	downward	0.2000	0.3000	623	0.2496	0.2921	0.0425
np025	15	downward	0.3000	0.4000	573	0.3494	0.3333	0.0161
np025	15	downward	0.4000	0.5000	518	0.4490	0.4517	0.0027
np025	15	downward	0.5000	0.6000	473	0.5493	0.6152	0.0659
np025	15	downward	0.6000	0.7000	623	0.6477	0.7368	0.0891
np025	15	downward	0.7000	0.8000	546	0.7487	0.8626	0.1140
np025	15	downward	0.8000	0.9000	357	0.8428	0.9524	0.1096
np025	15	downward	0.9000	1.0000	69	0.9239	0.9855	0.0616
np025	720	return	0.0000	0.1000	2900	0.0460	0.1155	0.0695
np025	720	return	0.1000	0.2000	1694	0.1484	0.2928	0.1444
np025	720	return	0.2000	0.3000	1375	0.2440	0.3105	0.0665
np025	720	return	0.3000	0.4000	684	0.3434	0.3246	0.0189

package	lead_min	target	bin_low	bin_high	n	mean_preds	std_rateabs	gap
np025	720	return	0.4000	0.5000	285	0.4443	0.4246	0.0198
np025	720	return	0.5000	0.6000	55	0.5423	0.3455	0.1968
np025	720	return	0.6000	0.7000	3	0.6079	0.6667	0.0588
np025	720	downward	0.0000	0.1000	1263	0.0618	0.1108	0.0490
np025	720	downward	0.1000	0.2000	1075	0.1480	0.2381	0.0901
np025	720	downward	0.2000	0.3000	1109	0.2495	0.3517	0.1021
np025	720	downward	0.3000	0.4000	885	0.3483	0.4395	0.0912
np025	720	downward	0.4000	0.5000	838	0.4475	0.4403	0.0071
np025	720	downward	0.5000	0.6000	840	0.5494	0.5131	0.0363
np025	720	downward	0.6000	0.7000	635	0.6490	0.6709	0.0219
np025	720	downward	0.7000	0.8000	258	0.7416	0.7403	0.0013
np025	720	downward	0.8000	0.9000	93	0.8409	0.6559	0.1849
np028	15	return	0.0000	0.1000	1700	0.0473	0.1047	0.0574
np028	15	return	0.1000	0.2000	892	0.1394	0.2085	0.0691
np028	15	return	0.2000	0.3000	465	0.2458	0.3011	0.0553
np028	15	return	0.3000	0.4000	368	0.3491	0.3234	0.0257
np028	15	return	0.4000	0.5000	253	0.4422	0.4862	0.0440
np028	15	return	0.5000	0.6000	87	0.5362	0.4253	0.1109
np028	15	return	0.6000	0.7000	7	0.6547	0.7143	0.0596
np028	15	downward	0.0000	0.1000	944	0.0399	0.0985	0.0586
np028	15	downward	0.1000	0.2000	720	0.1457	0.1708	0.0252
np028	15	downward	0.2000	0.3000	341	0.2413	0.2053	0.0361
np028	15	downward	0.3000	0.4000	191	0.3445	0.2618	0.0828
np028	15	downward	0.4000	0.5000	210	0.4473	0.3857	0.0616
np028	15	downward	0.5000	0.6000	202	0.5518	0.4109	0.1409
np028	15	downward	0.6000	0.7000	269	0.6503	0.5911	0.0593
np028	15	downward	0.7000	0.8000	393	0.7517	0.6412	0.1105
np028	15	downward	0.8000	0.9000	401	0.8450	0.7930	0.0520
np028	15	downward	0.9000	1.0000	101	0.9193	0.9802	0.0609
np028	720	return	0.0000	0.1000	1072	0.0556	0.1474	0.0918
np028	720	return	0.1000	0.2000	1299	0.1484	0.1440	0.0044
np028	720	return	0.2000	0.3000	647	0.2439	0.3091	0.0652
np028	720	return	0.3000	0.4000	390	0.3471	0.2667	0.0804
np028	720	return	0.4000	0.5000	296	0.4402	0.3446	0.0956
np028	720	return	0.5000	0.6000	63	0.5327	0.5079	0.0247
np028	720	return	0.6000	0.7000	5	0.6090	1.0000	0.3910
np028	720	downward	0.0000	0.1000	381	0.0599	0.2598	0.2000
np028	720	downward	0.1000	0.2000	551	0.1469	0.1706	0.0237
np028	720	downward	0.2000	0.3000	472	0.2497	0.2606	0.0109
np028	720	downward	0.3000	0.4000	480	0.3539	0.3979	0.0440
np028	720	downward	0.4000	0.5000	480	0.4551	0.2896	0.1655
np028	720	downward	0.5000	0.6000	615	0.5454	0.4618	0.0836
np028	720	downward	0.6000	0.7000	420	0.6460	0.4810	0.1650
np028	720	downward	0.7000	0.8000	280	0.7470	0.5214	0.2256

package	lead_min	target	bin_low	bin_high	m	mean_preds	rateabs	gap
np028	720	downward	0.8000	0.9000	93	0.8280	0.5376	0.2904
np025	15	downward	0.0000	0.1000	1820	0.0371	0.0193	0.0178
np025	15	downward	0.1000	0.2000	761	0.1453	0.1091	0.0362
np025	15	downward	0.2000	0.3000	503	0.2475	0.2445	0.0029
np025	15	downward	0.3000	0.4000	443	0.3487	0.3183	0.0304
np025	15	downward	0.4000	0.5000	381	0.4415	0.3963	0.0452
np025	15	downward	0.5000	0.6000	71	0.5378	0.6620	0.1241
np025	15	downward	0.6000	0.7000	17	0.6206	0.5882	0.0324
np025	720	downward	0.0000	0.1000	376	0.0781	0.0160	0.0621
np025	720	downward	0.1000	0.2000	3942	0.1598	0.0944	0.0655
np025	720	downward	0.2000	0.3000	1910	0.2372	0.0974	0.1398
np025	720	downward	0.3000	0.4000	563	0.3413	0.1012	0.2401
np025	720	downward	0.4000	0.5000	164	0.4345	0.1585	0.2760
np025	720	downward	0.5000	0.6000	40	0.5369	0.0250	0.5119
np025	720	downward	0.6000	0.7000	1	0.6144	0.0000	0.6144
np028	15	downward	0.0000	0.1000	2487	0.0603	0.0197	0.0406
np028	15	downward	0.1000	0.2000	404	0.1424	0.1064	0.0360
np028	15	downward	0.2000	0.3000	232	0.2502	0.1422	0.1079
np028	15	downward	0.3000	0.4000	261	0.3519	0.2222	0.1297
np028	15	downward	0.4000	0.5000	265	0.4493	0.2604	0.1890
np028	15	downward	0.5000	0.6000	113	0.5279	0.4159	0.1120
np028	15	downward	0.6000	0.7000	10	0.6153	0.5000	0.1153
np028	720	downward	0.0000	0.1000	81	0.0886	0.0741	0.0145
np028	720	downward	0.1000	0.2000	1598	0.1549	0.0582	0.0967
np028	720	downward	0.2000	0.3000	1570	0.2440	0.1057	0.1383
np028	720	downward	0.3000	0.4000	308	0.3348	0.0552	0.2796
np028	720	downward	0.4000	0.5000	183	0.4540	0.0874	0.3666
np028	720	downward	0.5000	0.6000	32	0.5294	0.1875	0.3419

The protocol and verification receipts bind the reliability rows to the frozen NP025/NP028 predictions.

S13. Joint first-passage baseline and route falsification (NP036)

NP036 fits a direct 17-class first-passage model with the joint feature arm under the same chronological windows and ExtraTrees budget. The route is evaluated both against this direct joint model and against the constructed joint distribution from the original joint event/timing heads. The direct baseline is a stronger end-to-end comparator; the constructed comparator isolates the value of replacing the conditional timing head while preserving the state head. The full six-lead matrix is stored in `outputs/next_paper/np036_joint_timing_baseline/joint_timing_baseline.csv`.

package	route_package	lead_time_min	metric	route_score	direct_joint_score	inttscore	grain_size	window_size	indefinite
np025	np026	15	joint_first	1.6759sage.1.6759BS	-	6996	nan		
					0.9672				
np025	np026	15	joint_first	1.6759sage.1.6759BS	-	6996	constructed_joint		
					0.3758				
np025	np026	60	joint_first	1.8884sage.1.8884BS	-	6996	nan		
					0.6992				
np025	np026	60	joint_first	1.8884sage.1.8884BS	-	6996	constructed_joint		
					0.3790				
np025	np026	120	joint_first	2.0093sage.2.0093BS	-	6996	nan		
					0.2344				
np025	np026	120	joint_first	2.0093sage.2.0093BS	0.1982	6996	constructed_joint		
np025	np026	240	joint_first	2.1434sage.2.1434BS	-	6996	nan		
					0.4094				
np025	np026	240	joint_first	2.1434sage.2.1434BS	1.0768	6996	constructed_joint		
np025	np026	480	joint_first	2.2158sage.2.2158BS	-	6996	nan		
					1.3754				
np025	np026	480	joint_first	2.2158sage.2.2158BS	1.1575	6996	constructed_joint		
np025	np026	720	joint_first	2.2317sage.2.2317BS	-	6996	nan		
					1.0744				
np025	np026	720	joint_first	2.2317sage.2.2317BS	-	6996	constructed_joint		
					0.0707				
np028	np029	15	joint_first	1.6883sage.1.6883BS	1.4172	3772	nan		
np028	np029	15	joint_first	1.6883sage.1.6883BS	1.5361	3772	constructed_joint		
np028	np029	60	joint_first	1.8668sage.1.8668BS	1.8379	3772	nan		
np028	np029	60	joint_first	1.8668sage.1.8668BS	1.1045	3772	constructed_joint		
np028	np029	120	joint_first	1.9608sage.1.9608BS	1.7399	3772	nan		
np028	np029	120	joint_first	1.9608sage.1.9608BS	0.6016	3772	constructed_joint		
np028	np029	240	joint_first	2.0436sage.2.0436BS	5.9699	3772	nan		
np028	np029	240	joint_first	2.0436sage.2.0436BS	0.4747	3772	constructed_joint		
np028	np029	480	joint_first	2.1908sage.2.1908BS	11.2023	3772	nan		
np028	np029	480	joint_first	2.1908sage.2.1908BS	0.7645	3772	constructed_joint		
np028	np029	720	joint_first	2.2855sage.2.2855BS	10.8416	3772	nan		
np028	np029	720	joint_first	2.2855sage.2.2855BS	0.6192	3772	constructed_joint		

S14. Weather-variable ablation (NP034)

NP034 compares calendar-only, thermodynamic-only, wind-vector-only and full-weather feature sets at 15, 240 and 720 min under the same tree budgets. The wind-vector channel supplies the positive state-risk contribution in both packages; thermodynamic-only features remain below the frequency reference in the tested ablation.

package	lead_minutes	dataset	task	metric	absolute_error	relative_error	wind_frequency_pct
np025	15	calendar	event	multiclass	0.184207	-	6996
						21.65830	
np025	15	calendar	downward	contin	0.17029	nan	2653
np025	15	wind_vector	event	multiclass	0.162075	10.31707	6996
np025	15	wind_vector	downward	contin	0.17694	nan	2653
np025	15	thermodynamic	event	multiclass	0.177281	-	6996
						11.65267	
np025	15	thermodynamic	downward	contin	0.18526	nan	2653
np025	15	full_weather	event	multiclass	0.162303	9.98776	6996
np025	15	full_weather	downward	contin	0.16972	nan	2653
np025	240	calendar	event	multiclass	0.183813	-	6996
						21.09426	
np025	240	calendar	downward	contin	0.17029	nan	2653
np025	240	wind_vector	event	multiclass	0.162015	10.39998	6996
np025	240	wind_vector	downward	contin	0.17694	nan	2653
np025	240	thermodynamic	event	multiclass	0.175693	-	6996
						9.36245	
np025	240	thermodynamic	downward	contin	0.18526	nan	2653
np025	240	full_weather	event	multiclass	0.162342	9.92732	6996
np025	240	full_weather	downward	contin	0.16972	nan	2653
np025	720	calendar	event	multiclass	0.184084	-	6996
						21.49735	
np025	720	calendar	downward	contin	0.17029	nan	2653
np025	720	wind_vector	event	multiclass	0.162064	10.32070	6996
np025	720	wind_vector	downward	contin	0.17694	nan	2653
np025	720	thermodynamic	event	multiclass	0.174896	-	6996
						8.21986	
np025	720	thermodynamic	downward	contin	0.18526	nan	2653
np025	720	full_weather	event	multiclass	0.162242	10.06424	6996
np025	720	full_weather	downward	contin	0.16972	nan	2653
np028	15	calendar	event	multiclass	0.184579	-	3772
						18.72647	
np028	15	calendar	downward	contin	0.16568	nan	1328
np028	15	wind_vector	event	multiclass	0.164427	5.07253	3772
np028	15	wind_vector	downward	contin	0.16869	nan	1328
np028	15	thermodynamic	event	multiclass	0.185863	-	3772
						26.51232	
np028	15	thermodynamic	downward	contin	0.17020	nan	1328
np028	15	full_weather	event	multiclass	0.167781	0.20442	3772
np028	15	full_weather	downward	contin	0.16936	nan	1328
np028	240	calendar	event	multiclass	0.184421	-	3772
						19.53029	
np028	240	calendar	downward	contin	0.16551	nan	1328
np028	240	wind_vector	event	multiclass	0.164438	5.49444	3772

package	lead_minutes	task	metric	absolute	relative	winfrequency_pct	
np028	240	wind_vector	downward	condition	0.16900	nan	1328
np028	240	thermodynamic	event	multiclass	0.186281	-	3772
					27.13345		
np028	240	thermodynamic	downward	condition	0.16953	nan	1328
np028	240	full_weather	event	multiclass	0.167059	1.19019	3772
np028	240	full_weather	downward	condition	0.16975	nan	1328
np028	720	calendar	event	multiclass	0.181675	-	3772
					20.37071		
np028	720	calendar	downward	condition	0.16737	nan	1328
np028	720	wind_vector	event	multiclass	0.163681	6.22230	3772
np028	720	wind_vector	downward	condition	0.16919	nan	1328
np028	720	thermodynamic	event	multiclass	0.186625	-	3772
					27.66617		
np028	720	thermodynamic	downward	condition	0.16990	nan	1328
np028	720	full_weather	event	multiclass	0.167578	0.40500	3772
np028	720	full_weather	downward	condition	0.16913	nan	1328

S15. Range and path supporting results (NP014)

NP014 supplies a separate amplitude and path-order channel. The first two rows compare the direct range arm with the old conditional range reference; the third compares the path-order assignment with the same range marginal after order shuffling. The table gives the exact headline values used in the main text; the full range, coverage and event-value matrices remain in the NP014 output package.

lead_min	task	candidate	reference	gain_pct	CI_low	CI_high	n
15	range	0.551	0.677	18.570	12.940	24.240	6059
	CRPS						
720	range	0.564	0.676	16.610	11.610	22.050	6059
	CRPS						
720	return	0.121	0.122	1.120	0.200	2.130	6496
	Brier;						
	shuf-						
	fled						
	order						

S16. Excursion-threshold sensitivity (NP037)

NP037 reconstructs the Suining aggregate from the archived SCADA and reruns the fixed power, weather and joint feature contracts at thresholds 0.15, 0.20 and 0.25. The 0.20 labels reproduce the frozen NP025/NP028 labels exactly for

every tested lead and package. At the two off-design thresholds, the weather-only state point gain remains positive at all six package-lead combinations; the timing table reports the added value of power history relative to weather.

S16.1 Weather state increment at thresholds 0.15 and 0.25

package	threshold	lead_minutes	relative_vs_weather_pct	power_added_to_weather_pct
np025	0.150	15	9.769	6996
np025	0.250	15	9.924	6996
np025	0.150	240	9.884	6996
np025	0.250	240	9.947	6996
np025	0.150	720	9.704	6996
np025	0.250	720	9.947	6996
np028	0.150	15	7.452	3772
np028	0.250	15	4.022	3772
np028	0.150	240	7.031	3772
np028	0.250	240	4.508	3772
np028	0.150	720	7.171	3772
np028	0.250	720	4.579	3772

S16.2 Power-history timing increment across thresholds

package	threshold	lead_minutes	power_added_to_weather_pct
np025	0.150	15	10.754
np025	0.150	240	0.833
np025	0.150	720	-0.131
np025	0.200	15	9.414
np025	0.200	240	2.496
np025	0.200	720	0.901
np025	0.250	15	6.387
np025	0.250	240	0.940
np025	0.250	720	1.164
np028	0.150	15	10.749
np028	0.150	240	1.583
np028	0.150	720	1.404
np028	0.200	15	8.923
np028	0.200	240	0.949
np028	0.200	720	4.337
np028	0.250	15	4.515
np028	0.250	240	-0.990
np028	0.250	720	2.868

S17. Mathematical synthesis and operational interface

The forecast family used in the paper is

$$y_{q,T} = \Phi_q(Y_T), \quad \widehat{F}_{q,h}^a = f_q(X_{T-h}^a, h), \quad \mathcal{L}_{q,h}^a = \mathbb{E} S_q(\widehat{F}_{q,h}^a, y_{q,T}).$$

The empirical information value is

$$\widehat{\text{IV}}_q(A \rightarrow B; h) = 100 \frac{\widehat{\mathcal{L}}_{q,h}^A - \widehat{\mathcal{L}}_{q,h}^B}{\widehat{\mathcal{L}}_{q,h}^A}.$$

For a downward occurrence probability $p_\downarrow$ and a conditional first-passage distribution q_τ , the no-crossing-inclusive accounting distribution is

$$\tilde{q}_k = p_\downarrow q_{\tau,k} \quad (k = 0, \dots, 15), \quad \tilde{q}_{16} = 1 - p_\downarrow.$$

The target-aligned route preserves the joint state vector and replaces only the conditional timing channel. Therefore, the constructed joint-score change is attributable to the timing-channel replacement under the fixed state identity; NP036 supplies the direct 17-bin joint comparison. The operational interface uses the return-event probability for a state alert and the conditional timing distribution for timing allocation. NP023 varies the alert threshold for a stated false-negative/false-positive loss while keeping forecast probabilities frozen.

S18. Capacity-matched information controls (NP038)

NP038 compares the original feature arms with compact weather summaries, a fixed projection, timestamp-matched noise, and power-plus-weather/noise arms at matched dimensions. Weather summaries use the four calendar columns plus mean, standard deviation, first node and last node for each issued weather variable. The projection is standardized on training rows before a fixed QR map; the noise arm is generated from timestamp-hashed random seeds. The complete 432-row score table, 648-row seed table, feature maps, labels, predictions and tree-complexity records are released in the NP038 directory.

package	comparison	min	max	mean
np025	power_weather_5670mary40_vs11329r_noise407.718	5670	11329	407.718
np025	weather_summary_25214vs_noise207.807	25214	207.807	26.768
np028	power_weather_6825mary40_vs9206r_noise408.074	6825	9206	408.074
np028	weather_summary_14278vs_noise205.915	14278	205.915	15.430

The rows below provide the primary matched state-score comparisons; interval bounds are paired seven-day blocks with 2,000 draws.

package	lead	minutes	comparison	candidate	reference	score	reduction	reduction	reduction	high	windows
np025	15		weather_0.608mary	2843vs_127.827	22.441	31.847	6996				
np025	15		power_weather_0.545mary	40.069	3.591	2.406	6996				
np025	60		weather_0.607mary	2834vs_127.825	21.769	31.244	6996				
np025	60		power_weather_0.569mary	40.670	2.942	2.433	6996				
np025	120		weather_0.607mary	2835vs_127.827	21.871	31.417	6996				
np025	120		power_weather_0.581mary	40.231	2.905	2.465	6996				
np025	240		weather_0.607mary	2829vs_126.525	21.336	30.785	6996				
np025	240		power_weather_0.603mary	40.338	2.489	2.908	6996				
np025	480		weather_0.608mary	2821vs_126.020	20.613	30.101	6996				
np025	480		power_weather_0.616mary	40.680	2.463	2.472	6996				
np025	720		weather_0.609mary	2818vs_125.524	20.274	29.715	6996				
np025	720		power_weather_0.617mary	40.329	2.419	2.398	6996				
np028	15		weather_0.685mary	2806vs_115.928	11.168	20.640	3772				
np028	15		power_weather_0.631mary	40.825	2.667	2.466	3772				
np028	60		weather_0.686mary	2811vs_115.320	11.543	21.271	3772				
np028	60		power_weather_0.665mary	40.763	3.004	2.659	3772				
np028	120		weather_0.689mary	2815vs_115.420	11.583	21.101	3772				
np028	120		power_weather_0.634mary	40.595	2.545	2.405	3772				
np028	240		weather_0.685mary	2812vs_115.626	11.768	21.649	3772				
np028	240		power_weather_0.654mary	40.856	2.632	2.574	3772				
np028	480		weather_0.688mary	2811vs_115.223	11.333	20.916	3772				
np028	480		power_weather_0.766mary	40.198	2.421	2.430	3772				
np028	720		weather_0.684mary	2813vs_115.925	12.211	21.106	3772				
np028	720		power_weather_0.777mary	40.206	2.972	2.429	3772				

NP038 also retains projection and timing comparisons whose signs vary by package or lead. These rows remain part of the release and are not promoted to the main narrative.

S19. All-lead route uncertainty and endpoint policy (NP039/NP040)

NP039 recomputes route, constructed-joint and direct-joint 17-bin products with a shared UTC 7-day block universe, 2,000 draws and fixed models. Conditional timing rows use the downward-event mask; state identity is checked against the frozen joint event head. NP040 first averages the six lead-wise ratios within each bootstrap draw and then takes the interval, avoiding interval averaging.

package	reference	metric	equal_weight	halmeq	gain	high	low	main	lead	interval	invalid	windows
np025	constructed-joint	2.548	median	794	ae	4.290	tes	-1.448	8.785	2653		
np025	constructed-joint	2.790	ps	1.011	4.891	-0.838	8.702	2653				

lead_minutes	cost_ratio	value	low	high	threshold	windows
720	5.000	41.985	32.622	48.650	0.140	6496
720	10.000	30.542	21.113	40.674	0.080	6496

S20.1 NP023 fusion comparison

lead_minutes	cost_ratio	candidate	reference	relative_cost	low	reduction	high
15	2.000	np013_selected	tpd14_selected	1.820	-11.368	13.994	
15	2.000	np014_selected	tpd14_selected	0.000	0.000	0.000	
15	2.000	np019_selected	tpd14_selected	0.590	-4.324	2.723	
15	2.000	np020_selected	tpd14_selected	0.590	-4.324	2.723	
15	5.000	np013_selected	tpd14_selected	0.849	-19.448	-1.383	
15	5.000	np014_selected	tpd14_selected	0.000	0.000	0.000	
15	5.000	np019_selected	tpd14_selected	2.664	-6.823	1.854	
15	5.000	np020_selected	tpd14_selected	2.664	-6.823	1.854	
15	10.000	np013_selected	tpd14_selected	2.851	-53.701	-13.898	
15	10.000	np014_selected	tpd14_selected	0.000	0.000	0.000	
15	10.000	np019_selected	tpd14_selected	1.610	-4.749	8.859	
15	10.000	np020_selected	tpd14_selected	1.610	-4.749	8.859	
720	2.000	np013_selected	tpd14_selected	8.264	-19.219	-0.791	
720	2.000	np014_selected	tpd14_selected	0.000	0.000	0.000	
720	2.000	np019_selected	tpd14_selected	0.875	-3.468	5.921	
720	2.000	np020_selected	tpd14_selected	0.875	-3.468	5.921	
720	5.000	np013_selected	tpd14_selected	10.403	-30.324	-10.886	
720	5.000	np014_selected	tpd14_selected	0.000	0.000	0.000	
720	5.000	np019_selected	tpd14_selected	3.402	-1.503	7.947	
720	5.000	np020_selected	tpd14_selected	3.402	-1.503	7.947	
720	10.000	np013_selected	tpd14_selected	12.334	-26.038	0.067	
720	10.000	np014_selected	tpd14_selected	0.000	0.000	0.000	
720	10.000	np019_selected	tpd14_selected	0.211	-5.054	4.787	
720	10.000	np020_selected	tpd14_selected	0.211	-5.054	4.787	

S21. Literature gap and anonymous reproduction

The verified literature matrix separates NWP ensemble ramps, joint time-distribution ramps, timing/intensity scoring, weather-regime conditioning and probabilistic calibration. The full comparison and official metadata are in temp/literature_gap_verified.md and literature_matrix_30plus.md. The public repository contains the processed aggregate/features/labels, frozen predictions, protocols, score tables, figure manifests and verification scripts. Raw SCADA, turbine-level coordinates, operational-status records and credentials are excluded. A reader can clone the repository anonymously and run the verification scripts without a GitHub account.